

LUAD Survival Analysis Project

Phase 3 — Clinical EDA & Baseline Model

Technical Summary for Sonica (Bioinformatics Lead)

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Project: Multi-modal machine learning for LUAD survival prediction using TCGA clinical data, RNA-seq gene expression, GTEx healthy lung reference, and immune microenvironment profiles.

This document: Covers Phase 3 — the complete clinical data pipeline from raw TCGA files to a working Cox proportional hazards baseline model with C-index = 0.70. This is the ML foundation that all subsequent models (DeepSurv, fusion) will be benchmarked against.

484	122	0.70	3
Patients in analysis	Deaths (events)	C-index (baseline)	Clinical features

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1. Data Source and File Structure

1.1 Source Dataset

Dataset	TCGA-LUAD (The Cancer Genome Atlas — Lung Adenocarcinoma)
Provider	GDAC Broad Institute (Firehose pipeline, 2016-01-28 release)
Access	Public — no login required
Original file	LUAD.Clinical.Level_1.2016012800.0.0.tar.gz

1.2 File Used

After extraction, the archive contains four clinical files. We use the **merged_only_clinical** file as it contains only core patient-level clinical variables (527 rows), free from biospecimen and admin metadata noise.

File name	LUAD.merged_only_clinical_clin_format.txt
Format	Tab-separated text (.txt), transposed — rows=variables, columns=patients
Raw dimensions	527 clinical variables x 522 patients
Missing values	Encoded as NA, [Not Available], or [Unknown]
Patient IDs	TCGA barcodes in format: tcga-05-4244

Why transposed? TCGA clinical files are stored with variables as rows and patients as columns — the opposite of the ML convention (rows=samples, columns=features). The first step in the pipeline is always to transpose this matrix.

2. Data Loading Pipeline

The loading pipeline runs in three steps, each producing a clearly named output:

Step	Action	Result Shape
1. Load raw file	<code>pd.read_csv</code> with <code>sep='\t'</code> , <code>index_col=0</code> , <code>na_values=['NA',...]</code>	527 rows x 522 cols
2. Transpose	<code>raw.T</code> — flip rows and columns	522 rows x 527 cols
3. Set index	<code>clinical.set_index('patient.bcr_patient_barcode')</code>	522 patients x 526 variables

Key loading parameter — `na_values`

TCGA files use three different strings to represent missing data. We tell pandas to treat all three as NaN on load so downstream operations work cleanly without manual string cleaning:

- NA
- [Not Available]
- [Unknown]

3. Survival Column Construction

3.1 Columns Extracted

From 527 clinical variables we extract exactly six columns needed for survival analysis:

Original Column Name	Renamed To	Role
patient.vital_status	vital_status	Dead or alive at last contact
patient.days_to_death	days_to_death	Days from diagnosis to death (dead patients only)
patient.days_to_last_followup	days_to_last_followup	Days tracked (alive/censored patients only)
patient.age_at_initial_pathologic_diagnosis	age	Age at diagnosis (continuous, years)
patient.gender	gender	Biological sex (male/female)
patient.stage_event.pathologic_stage	stage	TNM pathologic stage (I, II, III, IV)

3.2 Constructing survival_time and event

Key concept — Censoring: Some patients are still alive when the study ends. We don't know when they'll die — only that they survived at least X days. These patients are 'censored'. We cannot ignore them (that would discard 75% of our data and bias the model), so survival analysis handles them through the censoring mechanism.

Two new columns are constructed by combining the raw survival columns:

New Column	Logic	Code
survival_time	If dead: use days_to_death If alive: use days_to_last_followup	<code>days_to_death.fillna(days_to_last_followup)</code>
event	1 = died during study 0 = alive/censored at last contact	<code>(vital_status == 'dead').astype(int)</code>

4. Data Cleaning Decisions

All cleaning steps are documented below with the reasoning for each decision. Reproducibility requires that every dropped patient be explainable.

Step	Patients Removed	Reason	Remaining
Raw load	—	Starting point	522
Drop missing survival_time	19	Neither days_to_death nor days_to_last_followup present — no survival information available	503
Drop zero survival_time	19	survival_time = 0 days is biologically implausible and likely a data entry error	484
Final analysis dataset	—	All patients have valid survival_time > 0	484

Total data loss: 7.3% (38 of 522 patients removed). This is within acceptable range for TCGA clinical data. The 7.3% loss is documented and justifiable — both removal criteria are standard practice in published TCGA survival analyses.

5. Exploratory Data Analysis — Key Findings

5.1 Survival Time Distribution

Count (valid)	503 patients had valid survival time before zero-drop
Mean	543 days (~1.5 years)
Median	248 days (~8 months) — used as reference line in plot
Minimum	1 day (after zero removal)
Maximum	6,812 days (~18.7 years)
Std deviation	832 days — very high spread, indicating extreme variability in outcomes
Distribution	Heavily right-skewed — majority cluster under 500 days, long tail of survivors

Implication for modelling: The right-skewed, non-normal distribution means linear regression on survival time would be invalid. This is why we use survival analysis (Cox, DeepSurv) which makes no assumption about the shape of the survival time distribution.

5.2 Event Distribution (Censoring)

Dead (event=1)	122 patients — 25.2% of cohort
Censored (event=0)	362 patients — 74.8% of cohort
Implication	Three quarters of patients are censored — cannot use classification models
TCGA context	Typical event rate for TCGA cancer datasets; reflects long follow-up periods

5.3 Age at Diagnosis

Median age	66 years
Range	Approximately 35 to 89 years
Shape	Approximately bell-shaped, centred on 66
Biology	LUAD is predominantly a disease of older adults — very few patients under 45

5.4 Pathologic Stage Distribution

Stage I	~259 patients — dominant group
Stage II	~114 patients
Stage III	~78 patients

Stage IV	~25 patients — underrepresented
Bias note	TCGA over-represents early-stage resectable tumours because those patients had tissue available for sequencing. Stage IV is underrepresented. This limits generalisability to metastatic disease.

6. Kaplan-Meier Survival Curves

Kaplan-Meier (KM) curves show the probability that a patient is still alive at any given time point. The curve starts at 1.0 (100% alive) on day 0 and steps down each time a patient dies. Censored patients are marked with a tick but do not cause a step down. The shaded band is the 95% confidence interval — wider bands indicate fewer patients and higher uncertainty.

Three comparisons were plotted:

Comparison	Observation	Statistical Signal
Survival by Gender	Male patients (n=224) survive slightly longer than female (n=269). Curves statistically significant but confidence interval is wide.	Weak
Survival by Stage	Clear separation in expected order: Stage I > II > III > IV. Stage IV patients (n=25) died almost immediately.	Strong
Survival by Age Group	Younger patients (<65 years, n=231) survive longer than older (≥65 years, n=253). Separation is consistent.	Moderate

Key takeaway for Sonica: The KM curves validate that our clinical data is biologically coherent — the survival patterns match what is known from the LUAD literature. Stage is the dominant signal. These curves also set the expectation: any model that cannot beat this stage-based separation is not adding value.

7. Cox Proportional Hazards Model — Setup

7.1 What the Cox model does

The Cox proportional hazards model estimates how much each variable increases or decreases a patient's instantaneous risk of dying (the 'hazard'). It outputs a hazard ratio (HR) for each variable — a risk multiplier relative to a reference group.

HR > 1.0	Variable increases risk of death (worse prognosis)
HR < 1.0	Variable decreases risk of death (protective)
HR = 1.0	No effect on survival
p < 0.05	Effect is statistically significant
95% CI crosses 1.0	Effect is NOT statistically significant

7.2 Feature Encoding

The Cox model requires numeric inputs. Categorical variables are encoded as binary (0/1) dummy variables. One category is always left out as the reference group:

Feature	Encoding	Reference Group
age	Continuous (years) — no encoding needed	N/A (continuous)
gender	is_male: 1=male, 0=female	Female
stage	stage_II, stage_III, stage_IV (three binary columns)	Stage I

Note: We never create a column for the reference group — including it would cause perfect multicollinearity, making the model mathematically unsolvable.

8. Model Results and Interpretation

8.1 Full Model Summary

Variable	Coef	Hazard Ratio	95% CI	p-value	Significance
age	0.02	1.02	[1.00 – 1.04]	0.05	Borderline
is_male	-0.25	0.78	[0.54 – 1.13]	0.19	Not significant
stage_II	0.93	2.54	[1.61 – 4.02]	<0.005	★★★ Highly significant
stage_III	1.47	4.33	[2.74 – 6.86]	<0.005	★★★ Highly significant
stage_IV	1.30	3.67	[1.85 – 7.28]	<0.005	★★★ Highly significant

8.2 Variable-by-Variable Interpretation

- **Age (HR=1.02, p=0.05):** Each additional year of age increases death risk by 2%. Borderline significant. Compounds meaningfully — a patient 20 years older has ~1.49x higher risk.
- **Gender/is_male (HR=0.78, p=0.19):** Male patients show 22% lower risk than female, but this is NOT statistically significant (p=0.19, CI crosses 1.0). Gender is unlikely to be a useful feature in Phase 4.
- **Stage II (HR=2.54, p<0.005):** Stage II patients have 2.54x the risk of Stage I. Highly significant. Clinically expected.
- **Stage III (HR=4.33, p<0.005):** Strongest predictor. Stage III patients have 4.33x the risk of Stage I. This dominated the KM curves and it dominates the Cox model.
- **Stage IV (HR=3.67, p<0.005):** Significant but HR is slightly lower than Stage III — this is because Stage IV has only n=25 patients, making the estimate less stable (wide CI: 1.85 to 7.28).

8.3 Headline Result

C-index 0.7001	The model correctly ranks 70.0% of all patient pairs by survival risk.
	Published clinical-only Cox models on TCGA-LUAD typically report C-index 0.65–0.72. Our baseline is solidly within this range.

9. Outputs and File Locations

All outputs are committed to the shared GitHub repository under the `luad_survival` project:

File	Location	Contents
<code>clinical_survival.csv</code>	<code>data/processed/</code>	Cleaned survival dataframe: 484 patients x 10 columns. This is the primary input
<code>01_clinical_eda.png</code>	<code>outputs/figures/</code>	4-panel EDA figure: survival distribution, event counts, age distribution, stage distribution
<code>02_kaplan_meier.png</code>	<code>outputs/figures/</code>	3-panel KM curves: by gender, by stage, by age group.
<code>03_cox_hazard_ratios.png</code>	<code>outputs/figures/</code>	Forest plot of Cox model hazard ratios with 95% confidence intervals.
<code>cox_baseline_results.csv</code>	<code>outputs/results/</code>	Full Cox model summary: coefficients, hazard ratios, CIs, p-values for all variables
<code>01_clinical_eda.ipynb</code>	<code>notebooks/</code>	Complete reproducible notebook — all steps from raw file to Cox model.

`clinical_survival.csv` — Column Reference

Column	Type	Description
<code>vital_status</code>	string	Original: 'alive' or 'dead'
<code>days_to_death</code>	float	Days to death — NaN for alive patients
<code>days_to_last_followup</code>	float	Days to last contact — NaN for dead patients
<code>age</code>	float	Age at diagnosis in years
<code>gender</code>	string	'male' or 'female'
<code>stage</code>	string	Full pathologic stage label e.g. 'stage iiiia'
<code>survival_time</code>	float	PRIMARY: <code>days_to_death</code> OR <code>days_to_last_followup</code>
<code>event</code>	int	PRIMARY: 1=died, 0=alive/censored
<code>stage_group</code>	string	Simplified: 'Stage I', 'Stage II', 'Stage III', 'Stage IV'
<code>age_group</code>	category	'< 65 years' or '>= 65 years'

10. What Sonica Needs to Provide for Phase 4

Phase 4 (DeepSurv + multi-modal fusion) requires RNA-seq expression features and immune profiling features to be merged with the clinical survival labels in `clinical_survival.csv`. Parth's models are ready to receive these inputs — the pipeline is built to accept them. The items below are what the ML pipeline is waiting on.

Expression matrix

Expected path: `data/processed/expression_matrix.csv`

- TCGA-LUAD RNA-seq RSEM-normalised gene expression.
- Format: rows=patients (TCGA barcode, first 12 chars), columns=genes.
- Pre-processing: $\log_2(\text{RSEM}+1)$ normalisation, protein-coding genes only, top 5,000 most variable genes selected.
- Patient IDs must match `clinical_survival.csv` index exactly.

Dysregulation score matrix

Expected path: `data/processed/dysregulation_scores.csv`

- Per-patient per-gene z-scores vs GTEx healthy lung reference.
- Format: rows=patients, columns=genes (same gene set as expression matrix).
- Computed as: $(\text{tumor_expression} - \text{gtex_mean}) / \text{gtex_std}$ per gene.

Immune feature matrix

Expected path: `data/processed/immune_features.csv`

- CIBERSORT immune cell fractions + ssGSEA immune scores.
- Format: rows=patients, columns=immune features (22 CIBERSORT fractions + ssGSEA scores).
- All values should be on comparable scale (fractions sum to 1 for CIBERSORT).

Critical Matching Requirement

Patient ID format must match exactly. `clinical_survival.csv` uses lowercase TCGA barcodes with 12 characters: `tcga-05-4244`.

RNA-seq sample IDs in TCGA files are longer: `TCGA-05-4244-01A-11R-1757-07`. When exporting matrices from R, extract only the first 12 characters and convert to lowercase. In R: `substr(colnames(expr_matrix), 1, 12)` then `tolower()`. Parth will handle the inner join on the Python side — but the ID format must match.

Phase 3 Complete

Clinical data loaded and cleaned ✓
EDA completed with 4 publication-ready figures ✓
Kaplan-Meier curves plotted and interpreted ✓
Cox baseline model fitted: C-index = 0.70 ✓
All results committed to GitHub ✓

Next milestone: Phase 4 — RNA-seq loading, patient ID matching, DeepSurv implementation.